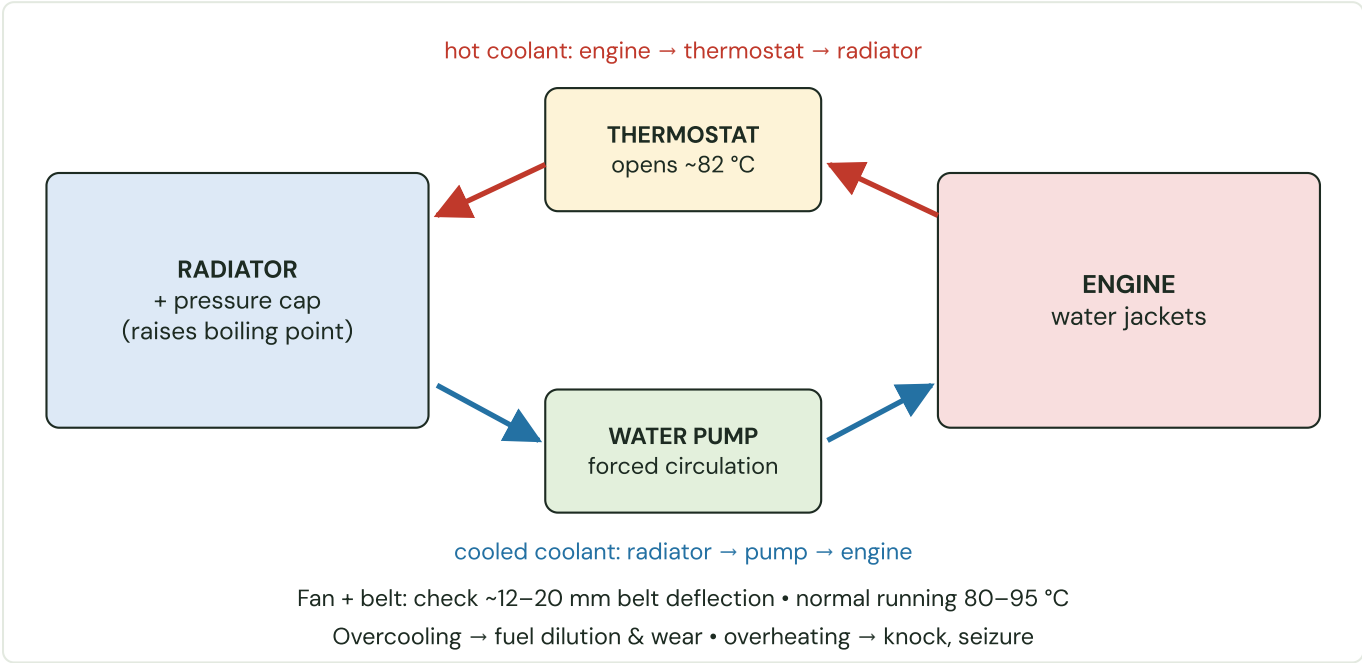
UNIT 4 · TOPIC 2(C)

Cooling and Lubrication Systems

Syllabus topic: Various systems of IC engines: operation, adjustment and troubleshooting

Cooling system



Forced-circulation water cooling (redrawn — the source figure showed the radiator-side flow reversed)

Type	Features
Air-cooled	Fins + blower — small engines, power tillers
THERMOSIPHON	Circulation by density difference, no pump — obsolete, slow
FORCED circulation (tractors)	Centrifugal pump, fan, radiator, thermostat, pressure cap

- **Thermostat:** begins opening ~**80–85 °C**; keeps the engine in the **80–95 °C** band.
- **Pressure cap:** raises the boiling point (~0.5–1.0 bar ⇒ +10–20 °C) and vents vacuum.
- **Coolant:** soft water + antifreeze/anticorrosive (or long-life coolant).

Overheating — check these causes:

- Low coolant or leaks.
- Loose fan belt (check mid-span deflection ~**12–20 mm**).
- Clogged radiator fins/core.
- Stuck thermostat.
- Late injection timing; overload.

Overcooling harms too: poor combustion, crankcase fuel dilution, corrosive condensate and higher wear. Remove the thermostat only in an emergency — never permanently.

Lubrication system

Functions: reduce friction and wear, cool, seal the rings, clean (carry deposits to the filter), cushion, protect from corrosion.

System	Where used
SPLASH (dipper on the con-rod)	Small engines
FORCED-FEED full-pressure	Tractor standard

Oil path in a full-pressure system:

1. Sump
2. Strainer
3. Gear/rotor pump
4. Relief valve
5. FULL-FLOW filter
6. Main gallery
7. Crank, cam and rocker bearings (oil cooler on larger engines)

- Normal oil pressure: ~**2–4 bar** at rated speed.

Oils

- **SAE** viscosity grades: SAE 30; multigrades such as 15W-40 span winter and summer.
- **API** service classes: diesel "C" series — CD / CF-4 / CI-4 …

Service and troubleshooting

- Check the oil level DAILY; change oil + filter at the maker's hours.
- **Low oil pressure:** low level, diluted oil, worn pump/bearings, stuck relief valve, choked strainer.
- **High oil consumption / BLUE smoke:** worn rings/liners, valve-guide seals.

Practice MCQs

Q56. The thermostat begins to open at about:

(a) 40 °C
(b) 60 °C
(c) 82 °C
(d) 100 °C

Q57. The radiator pressure cap:

(a) Cools the fan
(b) Raises coolant boiling point
(c) Filters coolant
(d) Drives the pump

Q58. Thermosiphon cooling works by:

(a) Pump pressure
(b) Density difference between hot and cold water
(c) Fan suction
(d) Thermostat

Q59. Overcooling an engine causes:

(a) Better economy
(b) Crankcase oil dilution, condensate corrosion and wear
(c) More power
(d) Nothing

Q60. Fan-belt tension is checked by mid-span deflection of about:

(a) 2–3 mm (b) 12–20 mm

(c) 50 mm (d) 100 mm

Q61. Tractor engines use which lubrication system?

(a) Petroil mixing (b) Splash only

(c) Forced-feed full-pressure (d) Gravity

Q62. Normal engine oil pressure at rated speed is about:

(a) 0.2–0.4 (b) 2–4

(c) 20–40 (d) 100 bar

Q63. 'SAE 15W-40' describes the oil's:

(a) API service class (b) Viscosity grade (multigrade)

(c) Flash point (d) Density

Answers

Q56 — (c) 82 °C

Q57 — (b) Raises coolant boiling point

Q58 — (b) Density difference between hot and cold water

Q59 — (b) Crankcase oil dilution, condensate corrosion and wear

Q60 — (b) 12–20 mm

Q61 — (c) Forced-feed full-pressure

Q62 — (b) 2–4

Q63 — (b) Viscosity grade (multigrade)

Sources: Liljedahl et al. — Tractors and their Power Units; Jagdishwar Sahay; manufacturer service practice; TREM notifications; NPTEL farm power lectures.